

MATH 1203 BIT & CIS DISCRETE MATHEMATICS

Lecture 5: GENERATING FUNCTIONS

Course Lecturer: M. KANYIKA

Applied Studies Department- MUST

13th March, 2026

Lesson Outcome

By the end of this lesson you should be able to:

- distinguish generating sequence and generating functions
- build generating functions
- solve generating functions using multiply shift and subtract technique
- solve recurrence relations with generating functions

INTRODUCTION

- I present to you fifth lecture of this course
- In the previous lecture, you learnt about the Recurrence relation.
- In this lecture, we will discuss about Generating functions.

Generating sequence and Series

Definition 1

a generating function as a formal power series that represents a sequence of numbers or objects.

Definition 2

A Power series is said to be infinite if it has infinite sum of terms of the form

$$\sum_{k=0}^{\infty} c_k x^k.$$

Generating sequence and Series CONT'D

- We expand

$$\sum_{k=0}^{\infty} c_k x^k \quad \text{as}$$

$$c_0 + c_1x + c_2x^2 + c_3x^3 + \dots$$

- We call such power series in generating function as a generating series.
- Generating series generates the sequence $c_0, c_1, c_2, \dots$ which is simply a sequence of coefficients of the infinite polynomial.

Generating sequence and Series CONT'D

Example 3

What sequence is represented by generating series $3 + 8x^2 + x^3 + \frac{x^5}{7} + 100x^6 \dots$

Solution:

- Since the coefficient of x^0 is 3, then $a_0 = 3$
- The coefficient of $x^1=0$, hence $a_1 = 0$
- $x^2=8$, then $a_2 = 8$
- Continuing we have $a_3 = 1$, $a_4 = 0$
- Similarly, $a_5 = \frac{1}{7}$ and $a_6 = 100$.
- Therefore the sequence is $3, 0, 8, 1, 0, \frac{1}{7}, 100, \dots$

Building Generating Function

- We can build a closed formula for generating functions from power series using multiply, shift and subtract technique

Example 4

Find the generating function of the following generating series.

$$1 + x + x^2 + x^3 \dots$$

Building Generating Function

Solution:

$$\begin{array}{rcl} S & = & 1 + x + x^2 + x^3 \\ -xS & = & x + x^2 + x^3 + \dots \\ \hline (1-x)S & = & 1 \\ S & = & \frac{1}{1-x} \end{array}$$

Therefore $1 + x + x^2 + x^3 = \frac{1}{1-x}$

Hence the generating function of $1, 1, 1, 1, \dots$ is $\frac{1}{1-x}$

Solving Generating Function

Differencing

- Generally as seen in the above example, we note that when differencing,
- each term is multiplied by its index and the entire sequence is shifted left one place.

Solving Generating Function

Example 5

Find the generating function of 1, 3, 5, 7, 9 ...

Solution: Let the generating function for 1, 3, 5, 7, 9, ... be A . Therefore we have

$$A = 1 + 3x + 5x^2 + 7x^3 + 9x^4$$

$$-xA = x + 3x^2 + 5x^3 + 7x^4 \dots$$

$$(1-x)A = 1 + 2x + 2x^2 + 2x^3 + 2x^4 \dots$$

$$(1-x)A = 1 + \frac{2x}{1-x}$$

$$A = \frac{1}{1-x} + \frac{2x}{(1-x)(1-x)}$$

$$A = \frac{1+x}{(1-x)^2}$$

Solving Recurrence Relations with generating functions

Example 6

The sequence 1,3,7,15,31,63,... satisfies the recurrence relation $a_n = 3a_{n-1} - 2a_{n-2}$. Find the generating function of the sequence.

Solution : Given that

$$a_n = 3a_{n-1} - 2a_{n-2}$$

$$a_n - 3a_{n-1} + 2a_{n-2} = 0$$

$$1 - 3x + 2x^2 = 0$$

Recurrence relation with generating functions Solution CONT'D

Lets call the generating function for the sequence by A.
Then the following holds

$$A = 1 + 3x + 7x^2 + 15x^3 + 31x^4 + \dots a_n x^n + \dots$$

$$-3xA = 0 - 3x - 9x^2 - 21x^3 - 45x^4 + \dots - 3a_n x^n + \dots$$

$$2x^2A = 0 + 0x + 2x^2 + 6x^3 + 14x^4 + \dots + 2a_n x^n + \dots$$

$$(1 - 3x + 2x^2)A = 1 + 0$$

$$A = \frac{1}{1 - 3x + 2x^2}$$

- The recurrence relation which gives the sequence 1, 3, 7, 15, 31, 63, ... has a generating function $\frac{1}{1-3x+2x^2}$.
- Lets simplify the value of A using partial decomposition

Partial Decomposition of recurrence relation with generating function

- Partial decomposition tells us that we can write this fraction as the sum of two fractions
- We start by factoring the denominator:

$$A = \frac{1}{1 - 3x + 2x^2} = \frac{1}{(1 - x)(1 - 2x)}.$$

- Then we decompose the given fraction

$$\frac{1}{(1 - x)(1 - 2x)} = \frac{a}{1 - x} + \frac{b}{1 - 2x}$$

for some constants a and b .

Partial Decomposition of recurrence relation CONT'D

- To find a and b, add two decomposed fractions using a common denominator. This gives

$$\frac{1}{(1-x)(1-2x)} = \frac{a(1-2x) + b(1-x)}{(1-x)(1-2x)}.$$

so

$$1 = a(1-2x) + b(1-x).$$

- This must be true for all values of x

Partial Decomposition of recurrence relation Involving generating functions

CONT'D

- If $x = 1$, then the equation becomes $1 = -a$ so $a = -1$. When $x = \frac{1}{2}$ we get $1 = b/2$ so $b = 2$. This tells us that we can decompose the fraction like this:

$$A = \frac{1}{1 - 3x + 2x^2} = \frac{1}{(1 - x)(1 - 2x)} = \frac{-1}{1 - x} + \frac{2}{1 - 2x}.$$

Hence

$$A = \frac{1}{1 - 3x + 2x^2} = \frac{-1}{1 - x} + \frac{2}{1 - 2x}.$$

This completes the partial fraction decomposition. Notice that these two fractions are generating functions we know.

THANK YOU FOR BEING GOOD AUDIENCE!